THE INDUSTRIAL STACK BEHIND ARTIFICIAL INTELLIGENCE

Where Capital Will Earn Superior Returns, Why the Market Misprices It, and How to Allocate

The largest AI spending pools will not necessarily generate the highest investment returns. Durable economic rent accumulates only where replication barriers rise faster than capital can overcome them — a narrow band the market systematically under-prices.

Economic Rent = Structural Capability × Economic Capture × Duration

IC Memorandum · 2026–2036
Prepared for portfolio managers, investment committee members, and senior technology investors.

Source tiers: T1 company filings / audited data; T2 recognized institutions (IEA, SEMI, Gartner, IDC, TrendForce, Counterpoint); T3 estimates and analytical judgment. Reported facts, normalized economics, and judgment are kept distinct. Figures are approximate and given as ranges where sources vary by quarter. Geography is treated as manufacturing concentration and supply-chain risk only. Forward statements are not forecasts; independent verification is required before transactional use.

SECTION 1

Executive Investment Summary

Core thesis. Capital is flowing disproportionately into AI infrastructure layers where the constraint is visible, but durable economic rent will accumulate only where replication barriers rise faster than capital can overcome them. The market is paying for visible scarcity; durable returns will accrue behind barriers capital cannot easily replicate.

Market misunderstanding. Consensus equates the largest spending pools with the largest rent pools, and treats the visible bottlenecks — power, grid, memory — as durable winners because scarcity confers pricing power today.

Variant perception. Scarcity that capital can relieve is not durable rent. The roughly $700bn of 2026 operator capital expenditure (T1) is weighted toward the most reproducible layers, where the capital cycle competes returns to the cost of capital; the narrow band earning durable rent absorbs a fraction of the spend.

Highest-conviction positions. Own the structural compounders — EUV lithography, the leading-edge foundry, and electronic design automation — and own the accelerator ecosystem with explicit valuation discipline. Among physical-layer positions, own the battery and storage leader only as a conditional exception.

Biggest risks. Custom-silicon migration against the accelerator franchise (nearest-dated, highest-severity); AI capex digestion compressing returns across the reproducible layers; and an accelerator valuation that already prices sustained scarcity economics.

What would change the thesis. Inference becoming hardware-agnostic at scale; a proprietary power-delivery standard with lock-in; AI value shifting downstream to integrators; or the China frontier ecosystem becoming self-sustaining.

Allocation conclusion. *Own the rising-barrier chokepoints, rent the cyclical scarcities to the capacity curve, avoid the capital traps regardless of revenue growth, and size the accelerator against its expectations rather than its quality.*

Why this matters for capital allocation: the most-funded layer and the best-returning layer are different layers; an allocation built around the visible constraint is built around the wrong layer.

SECTION 2

The Core Mispricing

The central error is treating the location of the constraint as the location of the rent. A bottleneck attracts capital; capital relieves scarcity; rent survives only where the barrier rises as the capital arrives; and most AI bottlenecks sit where the barrier does not rise.

The relief test. Can scarcity be relieved by capital, competition, substitution, or incumbent absorption? If it can, the rent is temporary regardless of how acute the shortage feels. Power, transformers, grid, and storage are the binding 2026 constraint and fail the test — each is relievable by capital, so returns normalize as capacity responds. The leading-edge chip fails in the opposite direction: no capital conjures a thirty-year lithography ecosystem, a foundry's tacit yield-learning, or a multi-million-developer software ecosystem, so the barrier rises with each node.

Layer group	Relief test	Economic outcome	Capital-allocation meaning
Power, grid, transformers, commodity infrastructure	Scarcity relievable by capital	Returns normalize toward cost of capital	Trade or avoid — not durable rent
Lithography, leading-edge foundry, EDA, ecosystem software	Scarcity not relievable; barrier rises	Durable excess returns survive	Own — structural rent concentrates here

Why this matters for capital allocation: conflating the visible constraint with the durable rent is the cycle's defining mis-allocation.

SECTION 3

Rent Durability Framework

Layers are ranked not by revenue growth but by **Economic Rent = Structural Capability × Economic Capture × Duration**, evaluating supply concentration, replication difficulty, switching cost, ecosystem dependence, qualification barriers, pricing power, ROIC durability, capital intensity, and substitution risk. The multiplication is the discipline: a layer must hold a barrier, convert it to returns, and sustain it; failure in any one collapses durable rent.

The question for each layer is why it earns excess returns and why competitors will fail to reproduce it. The answer is always replication difficulty — a lithography ecosystem cannot be assembled within the decade; an integrated rival cannot aggregate the demand that funds the foundry flywheel; the design flow cannot be switched mid-project; the accelerator software ecosystem is a decade-deep codebase.

Metric	Value	Period	Tier	Conf.	Capital-allocation meaning
EUV supply share	~100%	2025	T3	High	Structural monopoly
Leading foundry, pure-play share	~70% (from ~64% in 2024)	2025	T2	High	Near-monopoly at leading edge
EDA top-three combined share	~75% (~31/30/13)	2025	T2/T3	High	Flow-lock-in oligopoly
HBM suppliers	Three (shares vary by quarter)	2025	T2	Med	Oligopoly, not monopoly
2026 operator capex	~$700bn	2026	T1	High	Capital into reproducible layers

Reported facts (T1), institutional estimates (T2), and analytical judgment (T3) are kept strictly separate throughout this memorandum.

SECTION 4

Capital Allocation Classification

Category 1 — Structural Compounders

Rising barriers, durable pricing power, high ROIC, reinvestment advantage. **EUV lithography, the leading-edge foundry, EDA, and the accelerator ecosystem** (with valuation discipline). Owning through cycles works because the reinvestment is self-reinforcing — the more they earn, the further ahead they get, and additional capital *strengthens* the moat rather than inviting competition.

Category 2 — Cyclical Beneficiaries

Temporary scarcity, strong pricing today, capacity that eventually destroys returns. **HBM, transformers and grid equipment, the qualified power-device tier.** Trade, do not permanently own. **Buy signal:** a capacity shortage forming, lead times extending, capacity not yet responding. **Exit signal:** the capacity response appears — lead times normalize, large new capacity is announced, pricing peaks — because the capacity the high prices fund is precisely what ends the cycle.

Category 3 — Capital Traps

Strategic importance, revenue growth, weak economic capture. **Commodity solar, generic energy hardware, commodity wide-bandgap substrate, standalone cooling, overbuilt data-center capacity.** Avoid. Revenue growth does not equal shareholder return because reproducible capacity competes returns to the cost of capital; the trap is that the growth and the strategic importance are real while the returns are not.

SECTION 5

The Investment Case, Layer by Layer

Each position with business quality, market expectation, variant perception, evidence, and the expected-return driver — earnings growth, margin durability, multiple expansion, cycle timing, or market-misunderstanding correction — followed by what must happen for the investment to work and what breaks the thesis.

EUV lithography

Business quality: the cleanest in the stack — a monopoly toll on the leading edge. **Market expectation:** an equipment-cyclical multiple that under-prices the durability. **Variant perception:** this is not an equipment vendor; durability, not margin, is the asset, and the low headline margin (low-50s% gross, T1) misleads the market. **Evidence:** 100% of EUV (T3); ROIC ~15 points above cost of capital, record backlog (T1). **Return driver:** margin durability plus earnings compounding. **Works if** the market recognizes the durability and High-NA ramps; **breaks if** a substitute patterning technology reaches scale.

Leading-edge foundry

Business quality: exceptional. **Market expectation:** a geopolitical-concentration discount. **Variant perception:** a near-monopoly with rising ASPs, not a commodity. **Evidence:** ~70% pure-play share (T2); ~60%/51% gross/operating margins, ~35% ROE (T1). **Return driver:** earnings growth plus multiple expansion if the discount narrows. **Works if** the 2nm monopoly is sustained; **breaks if** demand aggregation fragments.

Accelerator ecosystem

Business quality: the highest current quality in AI. **Market expectation:** continued scarcity economics and durable lock-in are priced in. **Variant perception:** the gap between business quality and embedded expectations is the entire investment question. **Evidence:** ~75% gross margin and ~75% ROIC (T1), above the historical mid-50s-to-mid-60s range; a real and nearest-dated ASIC threat. **Return driver:** whether earnings beat already-embedded expectations — not clean compounding. **Works if** the ecosystem captures inference and enterprise beyond what is priced; **breaks if** inference goes hardware-agnostic. A great company; a great investment only at a price that leaves room.

Electronic design automation

Business quality: a compounder hiding in plain sight. **Market expectation:** a mature, low-growth software multiple. **Variant perception:** the flow lock-in and the AI-design-complexity tailwind are under-rated. **Evidence:** ~75% combined share, ~80%+ gross margins, ~30%+ R&D; (T1/T2). **Return driver:** earnings growth plus a re-rating. **Works if** AI design complexity deepens the lock-in; **breaks if** an open or integrated flow reaches advanced-node parity (low probability).

HBM (high-bandwidth memory)

Business quality: cyclical, not structural. **Market expectation (currently):** a structural re-rating the cycle will not sustain. **Variant perception:** it is still memory, still a three-supplier oligopoly. **Evidence:** peak margins ~60% (T3); a +275–300% DRAM price forecast for 2025–2027 (T3). **Return driver:** cycle timing only. **Works if** capacity discipline holds; **breaks** on classic oversupply.

Grid equipment

Business quality: moderate. **Market expectation (currently):** a durable secular re-rating of a cyclical industry. **Variant perception:** a relievable super-cycle. **Evidence:** ~4-year transformer lead times (T2/T3), multiple producers. **Return driver:** cycle timing. **Works if** the super-cycle outruns the capacity response; **breaks** when lead times normalize.

Battery and storage leader

Business quality: strong in storage. **Market expectation:** oscillates between EV-cyclicality and a scale-moat. **Variant perception:** the rent is the qualification-gated storage segment, not scale. **Evidence:** ~17% net, ~27% storage gross margin (T1). **Return driver:** earnings growth in storage plus EV-cycle timing. **Works if** storage demand grows and the moat holds; **breaks** on chemistry commoditization.

SECTION 6

Expected Return Framework

The table separates whether an investment works because the company is excellent, the stock is mispriced, or the cycle is favorable — different propositions requiring different holding behavior.

Layer	Business quality	Valuation risk	Expected return driver	Key risk	What must happen for upside	What breaks the thesis
EUV lithography	Exceptional	Low–Med	Margin durability + compounding	Alternative lithography	Durability recognized; High-NA ramps	Substitute patterning at scale
Leading-edge foundry	Exceptional	Low–Med	Earnings growth + discount narrowing	Geo / second-sourcing	2nm monopoly sustained; discount narrows	Demand aggregation fragments
Accelerator ecosystem	Exceptional (current)	High	Earnings beating embedded expectations	ASIC migration / margin normalization	Ecosystem captures inference & enterprise	Inference goes hardware-agnostic
EDA	Very strong	Low–Med	Earnings growth + re-rating	Open-flow parity	AI design complexity deepens lock-in	Full-flow alternative at parity
HBM	Cyclical	High at peak	Cycle timing	Oversupply	Capacity discipline holds	Classic oversupply
Grid equipment	Moderate	High at peak	Cycle timing	Lead-time normalization	Super-cycle outruns capacity	Lead times normalize

Layer	Business quality	Valuation risk	Expected return driver	Key risk	What must happen for upside	What breaks the thesis
Battery / storage	Strong (storage)	Medium	Earnings growth + cycle	Chemistry com moditization	Storage demand grows; moat holds	Chemistry shift erases edge

The accelerator ecosystem is the highest-quality business in AI and the position with the highest expectations risk. The return driver dictates holding behavior: margin-durability compounders are owned through cycles; the discount-narrowing name is owned with the catalyst in view; the expectations-dependent name is owned only at a price that leaves room; cycle-timing names are rented and exited on the capacity signal. Mistaking a cycle-timing return for a compounding one is the most expensive error in the table.

SECTION 7

Capital Cycle Analysis — the Empirical Record

The discriminating question across every major capital cycle is whether additional capital strengthens the moat or destroys returns. Five cases answer it.

Historical case	Did capital strengthen the moat or destroy returns?	What earned the rent
Electricity infrastructure	Destroyed returns — reproducible generation became a regulated utility	Equipment patents; standards; enabled industries
Internet infrastructure	Destroyed returns — fiber/transport overbuilt to commodity/negative returns	Asset-light platforms with network effects
Solar manufacturing	Destroyed returns — dominant capacity, ROIC destruction, losses below cost	Almost nothing (commodity); scale ≠ rent
Memory semiconductor	Destroyed returns cyclically — up-cycle capacity collapses margins	No durable rent; scarcity ≠ structural rent
Semiconductor equipment	Strengthened the moat — IP and qualification barriers persist through cycles	The tool oligopoly; replication difficulty

Applied to AI: capital into the chokepoints — lithography, leading-edge foundry, design tools — strengthens the moat, as in equipment. Capital into the reproducible physical layers — power, grid, generic devices, commodity capacity — destroys returns, as in electricity, internet, and solar. The single test of whether capital strengthens or destroys is the test of where to allocate, grounded in the empirical record rather than a forecast.

SECTION 8

Scenario Sensitivity

The rent-durability scores under base, bear, and bull cases, with the single variable that moves each. Scores are calibrated judgment (T3), not measurement; the structural classifications are firmer than the duration estimates.

Layer	Base	Bear	Bull	Single variable that changes the conclusion
EUV lithography	8.1	7.0	8.5	Substitute existence / served-market size
Leading-edge foundry	6.0	4.5	6.8	Demand-aggregation integrity
Design tools (EDA)	5.8	4.5	6.5	Full-flow alternative at parity
Accelerator ecosystem	5.2	3.0	6.5	Inference portability
Battery / storage leader	2.5	1.5	3.5	Chemistry shift

Layer	Base	Bear	Bull	Single variable that changes the conclusion
HBM memory	1.5	0.8	3.0	Capacity discipline
Grid equipment	1.0	0.6	2.0	Lead-time normalization speed

Stable compounders versus single-variable bets. Lithography and foundry hold their scores across scenarios because their swing variables are low-probability and long-dated. The accelerator carries the widest base-to-bear range (5.2 down to 3.0): a single variable — inference portability — moves it from a durable oligopoly to contested.

SECTION 9

Falsification Dashboard

An IC monitoring system: each thesis as current belief, supporting evidence, the evidence that would invalidate it, and the action if invalidated.

Thesis	Supporting evidence	Invalidating evidence	Action if invalidated
Accelerator: ecosystem creates durable pricing power	Lock-in; ~75% margins; installed base (T1)	Inference hardware-agnostic; ASIC share rises; margin compression	Reduce exposure
Rent concentrates upstream	Monopoly/oligopoly economics; rising barriers (T1/T2)	Value shifts downstream to integrators	Rotate toward integrators
Power infrastructure commoditizes	Reproducible capacity; substrate insolvency; utility returns (T1/T2)	Proprietary power-delivery standard forms	Reclassify HVDC toward ownership
HBM is cyclical, not structural	Three-supplier oligopoly; cyclical margin history (T2/T3)	Capacity discipline holds through downcycle	Re-rate toward ownership
Grid super-cycle is temporary	Multiple producers; relievable capacity (T2)	Lead times keep extending despite new capacity	Extend cyclical hold; reassess
EUV monopoly is durable	100% supply; no commercial alternative (T3)	Alternative patterning reaches commercial scale	Reduce highest-conviction position
China is frontier-constrained	HBM, lithography, EDA, yield gaps (T2/T3)	Domestic ecosystem becomes self-sustaining	Reduce assumed durability of leading-edge rent

SECTION 10

Valuation Discipline

Identifying great businesses is not identifying great investments. **A great company can be a poor investment when expectations are too high.** For each winner: what is priced in, what assumption creates downside, what creates upside.

Lithography is priced as a cyclical equipment supplier, so the monopoly durability is arguably under-priced — downside requires a severe sustained export restriction; upside requires the market to recognize the monopoly. **The foundry** carries a geopolitical discount that may over-price the tail — downside is geographic disruption; upside is the discount narrowing. **The accelerator** is the sharpest case: it may remain the highest-quality business in AI while delivering ordinary equity returns, because the price already embeds sustained demand and elevated margins — downside is ASIC migration normalizing margins; upside requires the ecosystem to capture inference and enterprise beyond what is priced. The discipline: use this analysis as the quality screen and apply a valuation overlay before sizing.

SECTION 11

Investment Decision Framework

Layer	View	Confidence	Action	Key trigger
EUV lithography	Structural monopoly	High	Long-term ownership	Alternative lithography
Leading-edge foundry	Durable oligopoly	High	Long-term ownership	Customer second-sourcing
Design tools (EDA)	Flow-lock-in oligopoly	High	Long-term ownership	Open/integrated flow at parity
Accelerator ecosystem	Durable oligopoly, lower duration	Medium	Own w/ valuation discipline	ASIC migration / abstraction
Battery / storage leader	Conditional exception	Medium	Selective ownership	Chemistry commoditization
Semiconductor equipment	Durable oligopoly (lower)	Medium	Own / cyclical	Capex downturn
HBM memory	Cyclical scarcity	Medium	Cycle exposure / trade	Capacity discipline
Grid equipment	Temporary shortage	Medium	Tactical only	Lead-time normalization
Advanced packaging	Incumbent-held	Medium	Monitor	Independent standard
HVDC / optical interconnect	Ownership uncertain	Low	Monitor	Standard formation
Commodity infrastructure	Scale without rent	High	Avoid	—

Allocation conclusion. Weight to the structural compounders, with the accelerator sized against its valuation and its base-to-bear scoring range; hold the cyclical beneficiaries only with an exit discipline tied to the capacity signal; hold none of the capital traps for revenue growth. The core thesis stands — AI physical infrastructure follows the electricity, internet, and solar pattern of huge investment, strategic importance, and eventual return compression, while the AI semiconductor chokepoints follow the semiconductor pattern of rising barriers, ecosystem lock-in, and durable rent. The master indicator is the layer-by-layer ratio of AI revenue to AI capital expenditure.

APPENDIX

Methodology, Financials, and Source Discipline

A. Economic Rent scoring methodology

Economic Rent = Structural Capability × Economic Capture × Duration, each a weighted 1–10 composite, the product scaled to 0–10. Multiplication enforces that all three must hold; failure in any one collapses durable rent. Structural Capability and Economic Capture are anchored in observed data; Duration is primarily analyst judgment and carries the widest ranges. Economic Capture is scored on normalized (through-cycle) economics for cyclical layers, not peak.

Structural Capability	w	Economic Capture	w	Duration	w
Market concentration	25%	ROIC	30%	Replacement difficulty	25%
Replication difficulty	25%	Gross margin	20%	Switching cost	25%
Technology moat	20%	Operating margin	20%	Transition resistance	20%
Qualification barriers	15%	Pricing power	20%	Customer-dependence resist.	15%
Ecosystem dependence	15%	Cash conversion	10%	Capital barrier	15%

B. Financial comparison (approximate, FY2024–25)

Layer	Gross margin	Operating / net	ROIC reported	ROIC normalized	Tier / Conf.
Lithography	low-50s%	OM low-30s%	~mid-20s%	similar	T1 / High
Leading foundry	~60%	OM ~51%, ROE ~35%	~20–32%	~mid-50s% mid-cycle	T1 / High
Accelerator	~75%	net ~56%	~75% reported	lower, lock-in-dependent	T1 rep. / Med norm.
EDA	~80%+	high	high, stable	similar (recurring)	T1 / High
HBM	~60% peak	cyclical	cyclical peak	ordinary through-cycle	T3 / Med
Battery leader	~24% (27% storage)	net ~17%	solid	commoditization-exposed	T1 / High
Semiconductor equipment	~45–48%	high	high, persistent	persistent (IP barrier)	T1 / High

C. Source discipline

T1 — company filings and audited data. **T2 —** recognized institutions (IEA for power demand; TrendForce and Counterpoint for foundry, DRAM, and HBM shares; SEMI for equipment; Gartner and IDC for market structure). **T3 —** market estimates and analytical judgment (EUV-supply and installed-base percentages; HBM peak-margin figures; the DRAM price-cycle forecast; the battery cost curve; duration scores; scenario probabilities; migration paths). These categories are never mixed. Confidence is highest for current structure and lowest for future rent migration.

This memorandum is a rent-durability and capital-allocation assessment, not investment advice. Numbers are current to Q2 2026, tagged by source tier and confidence, and given as ranges where sources vary by quarter; reported, normalized, and judgment figures are distinguished. Geography is treated as manufacturing concentration and supply-chain risk only. Forward statements are not forecasts. Independent verification is required before any transactional use.